

ASSESSMENT TOOL-1

Course code	CSA 0702	Course Title	Computer network
CO Assessed	CO1	Assessment	Tool - Short Answer
Weightage	40%.	Total mark)	20 marks

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Total mark)	20	Obtained marks	17
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Result
20/7/26

1) A company needs to 150 MB design file from current
with bandwidth of 50 Mbps.

Answer:-

File size = 150 MB

Bandwidth = 50 Mbps

Segment size = 1500 bytes

Frame overhead = 40 bytes

Calculation:

File size = 150 MB

Bandwidth = 50 Mbps

$$150 \times 1024 \times 1024 = 15712861400 \text{ bytes}$$

Number of segment

$$15712861400 \div 1500 = 10481907 \text{ segment}$$

frames transmitted = 10481907

over head

$$10481907 \times 40 = 419276280 \text{ bytes}$$

transmission time:

$$150 \text{ MB} = 1200 \text{ mb}$$

$$1200 \div 50 = 24 \text{ seconds}$$

final answer:

$$\text{Number of segment} = 104858$$

$$\text{Number of frame} = 104858$$

$$\text{total overhead} = 4,194,320 \text{ bytes}$$

$$\text{transmission time} = 24 \text{ seconds}$$

Conclusion:

The transport layer divides large file into smaller segments and provides reliable delivery through acknowledgment.

2) The transport layer protocol uses a sliding window size of 6 frames.

Given:

$$\text{Window size} = 6$$

$$\text{frame} = 48$$

Calculation:

Windows required

$$48 \div 6 = 8 \text{ windows}$$

$$1 \text{ data frame} \times 8 \text{ windows} = 8 \text{ transmission.}$$

Theory:

Flow control ensures that sender transmits data only as fast as the receiver can process it. If it doesn't, receiver buffer overflows, minimizing packet loss.

3)

A 12,000 bytes IP packet is transmitted through router:

Given:

$$\text{packet size} = 12,000 \text{ bytes}$$

$$\text{MTU} = 1500 \text{ bytes}$$

$$\text{Header} = 20 \text{ bytes}$$

Calculation:

payload per fragment

$$(1500 - 20 = 1480 \text{ bytes})$$

Fragment):

$$1200 \div 1480 = 0.81$$

$$= 9 \text{ fragments}$$

Header overhead

$$9 \times 20 = 180 \text{ bytes}$$

Final answer:

$$\text{Fragment) = 9}$$

$$\text{Header overhead} = 180 \text{ bytes}$$

4)

when

calculation:

$$\text{window size} = 6$$

$$\text{Frame} = 48$$

$$\text{window} = 6$$

$$\text{Retransmission} = 8$$

Final conclusion:

Flow control balances the transmission speed between sender and receiver, preventing congestion and ensuring efficient communication.

5) A multimedia application transmits a 600 MB video file.

Soln:

Given:

File size = 600 MB

Check point = every 60 MB

Failure = 390 MB

Calculation:

check point

$390 \div 60 = 6$ (check point)

last check point = 360 MB

Retransmission:

$$390 - 360 = 30 \text{ MB}$$

Final answer:

$$\text{Check point} = 6$$

$$\text{Retransmitted data} = 30 \text{ MB}$$

The session layer create - synchronization checkpoint during communication. If transmission fails,

6) Before transmission the presentation layer compress 900 MB file by 40%.

Given

Calculation:

Compressed size

$$900 \times 60\% = 540 \text{ MB}$$

Encrypted size

$$540 \times 105\% = 567 \text{ MB}$$

Reduction:

$$(900 - 567) \div 900 \times 100 = 37\%$$

Final Answer:

compressed file = 540 MB

Encrypted file = 567 MB

Reduction = 37%

The presentation layer compresses data to reduce transmission time and encrypts to protect confidentiality and prevent unauthorized access.

7)

An FTP application transfers 3 MB of data over a network with an effective throughput of 120 Mbps.

Answer

given

$$3 \text{ MB} = 3072 \text{ MB}$$

Request

$$3072 \div 3 = 1024$$

transmission on time

$$3072 \div 3 = 1024$$

Transmission time

$$3072 \text{ MB} = 24576 \text{ mb}$$

$$24576 \div 120 = 204.8 \text{ seconds}$$

Final answer:

$$\text{Request} = 1024$$

$$\text{Transmission time} = 204.8 \text{ seconds}$$

The application layer provides services such as file transfer (FTP), email, web browsing.

8)

A packet and experience processing delays at different OSI layers.

given:

calculation:

processing delay

$$4 + 3 + 0.12 + 1 = 5.5 \text{ ms}$$

propagation delay = 18ms

transmission delay = 12ms

total delay:

$$5 + 18 + 12 = 45 \text{ ms}$$

final answer:

$$\text{total delay} = 45 \text{ ms}$$

Each byte delays process data before transmission

9). A network transmits 50 ms of useful data using 1600 byte frames. Each frame contains 20 bytes of protocol overhead.

Answer:

Calculation:

payload per frame

$$1600 - 20 = 1580 \text{ bytes}$$

Frame:

$$8318861080 \pm 1520285180$$

Overhead

$$55108 \times 80 = 4,415,140 \text{ bytes}$$

Efficiency:

$$(1820 \pm 1600) \times 100 = 95\%$$

Final Answer:

$$\text{Frame} = 55,108$$

$$\text{Overhead} = 4,415,140 \text{ bytes}$$

$$\text{Efficiency} = 95\%$$

protocol overhead (conflict) header and trailers
added network layer

Answer

given:

Calculation:

Compressed file

$$240 \times 70\% = 168 \text{ MB}$$

Compressed bytes

$$168 \times 1024 \times 1024 = 176,160,768 \text{ bytes}$$

Segment 1

$$176,160,768 \pm 1400 = 125,1829$$

Protocol overhead

$$125,1829 \times 60 = 7,549,740 \text{ bytes}$$

Final transmitted data

$$176,160,768 + 7,549,740 = 183,710,508 \text{ bytes}$$

($\approx 175.2 \text{ MB}$)

Final Answer:

$$\text{Compressed file} = 168 \text{ MB}$$

$$\text{Segment} = 125,1829$$

$$\text{Protocol overhead} = 7,549,740 \text{ bytes}$$